

Metadata Creation, Transformation and Discovery for Social Science Data Management: The DAMES Project Infrastructure

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**UNIVERSITY OF
STIRLING**



**University
of Glasgow**



**National
e-Science
Centre**



DAMES: Background

- **DAMES: Case studies, provision and support for data management in the social sciences**
- **A 3 year ESRC research “node”**
 - **February 2008-February 2011**
 - **Funded by NCeSS**
- **A cross subject collaboration:
Social Science and Computing Science**
- **A cross institutional collaboration:**
 - **Applied Social Science and Computing Science at Stirling University**
 - **National eScience Centre at Glasgow University (specialising in e-Health data and security)**

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- Driven by social science needs for both introductory and advanced ***data management operations***
 - “In practice, social researchers often spend more time on data management than any other part of the research process” (Lambert)
 - A ‘methodology’ of data management is relevant to social science literatures on ‘harmonisation’, ‘comparability’, ‘reproducibility’
 - Why is **e-Social Science** relevant?
 - Dealing with distributed, complex, heterogeneous data
 - Generic data requirements / provisions
 - Exploiting “virtual organization” concepts

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DAMES: Themes

- Four social science themes
 - Grid Enabled Specialist Data Environments: occupations; education; ethnicity
 - Micro-simulation on social care data
 - Linking e-Health and social science databases
 - Training and interfaces for data management support
- Underlying computer science research themes
 - Metadata
 - Data abstraction
 - Data fusion
 - Workflow modelling
 - Data security

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The GEODE project

- DAMES builds on previous work on GEODE project
 - *Grid Enabled Occupational Data Environment* (Gayle, Lambert, Tan, Turner)
- GEODE:
 - A virtual community of occupational information researchers
 - Gateway for occupational information (web portal)
 - Data abstraction (format independence)
 - Uniform access to resources
- Occupational data curation
 - Annotation of data using DDI 2



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GridSphere Portal - Microsoft Internet Explorer provided by University of Stirling

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Print Mail

Address http://geode.cs.stir.ac.uk:8080/gridsphere/gridsphere?cid=114 Go

Grid Enabled Occupational Data Environment

Logout
Welcome, Paul Lambert

Welcome Administration Feedback Deposit data **GEODE Search** GEODE Browse Link data resources

Search Data Resources

Portlet for searching data resources (G1 and deposited)

Resource ID	Resource Title	Abstract	Supplier	Supplied Date	Original Creator	Original Production Date	Affiliation	Email	Occupational Classifications Us
127	CAMSIS for Russia	View Abstract	Paul Lambert	2007-11-20	Paul Lambert; Ken Prandy	2003	University of Stirling	pl3@stirling.ac.uk	isco-88 (International Standard Classification of Occupations 1988)

Results of index search

Title	Authors	Countries	Time Periods	Service URI	Resource ID	Corresponding Uncurated Resources	View DDI	View Occupational Matching Log
CAMSIS scales for Russia using ISCO-88	Paul Lambert	Russian Federation	2001	http://139.153.254.158:8888/wsrf/services/geode/GEODEFactoryService	paullambert-camsisrussia	127	View DDI	View Occupational Matching L

Hit Enter or click Find Resources

[Help on search](#)

Search:

[Find Resources](#)

search results.iso

Done Local intranet

start P. I.. L.. p. C. N. C. s.. G. C. H.

10:58

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DAMES goals

- Building on GEODE experience and technology:
- Build an integrated DAMES ***Grid-based portal***:
 - Combining occupational, ethnicity, educational, health data resources: “GE*DE
 - Allowing researchers to deposit* and search heterogeneous data resources (**curation tool*)
 - Providing online services/‘tools’ that enable researchers to carry out repeatable and challenging data management techniques (such as fusion, matching, imputation...)
- *Facilitating access* is an important goal

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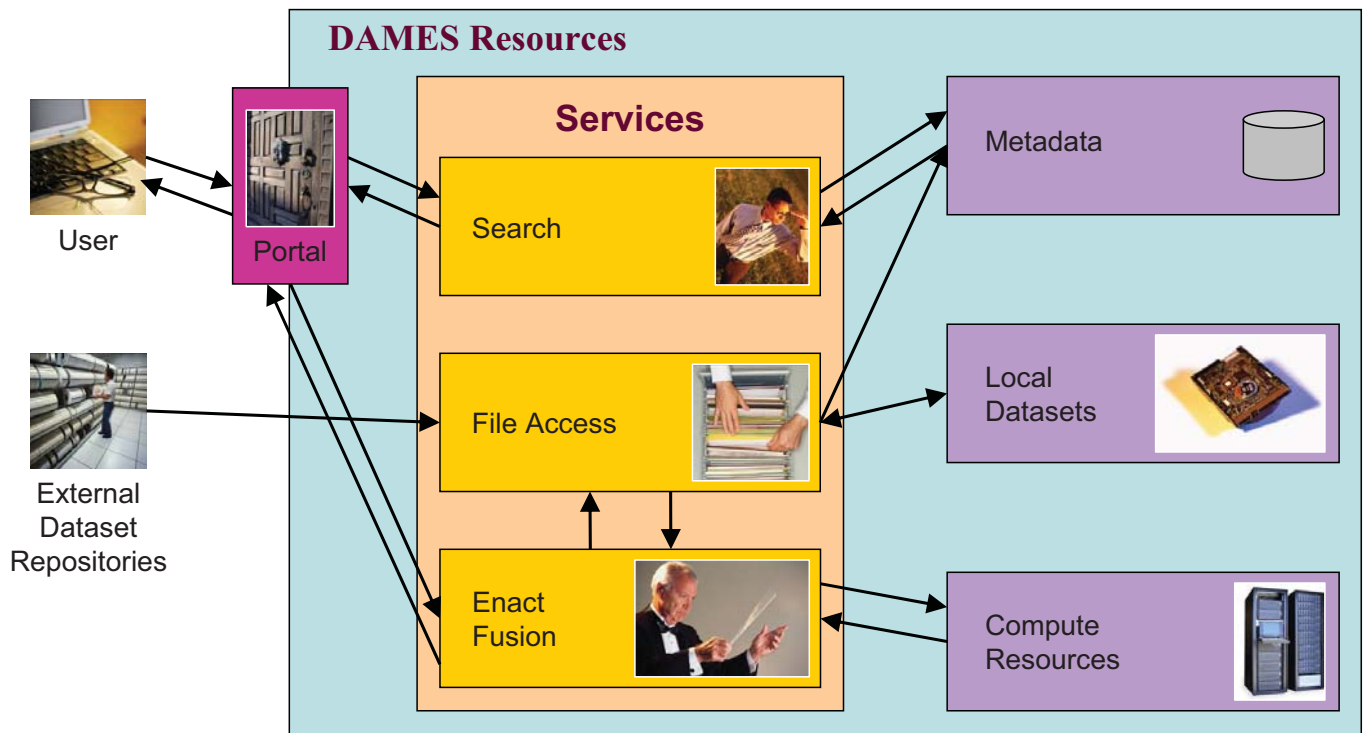
DAMES scenarios

- Curation scenarios could include:
 - Uploading occupational data to distribute across academic community
 - Recording data properties prior to undertaking data fusion involving a survey and an aggregate dataset
- Fusion scenarios could include:
 - Linking a micro-social survey with aggregate occupational information (deterministic link)
 - Enhancing a survey dataset with ‘nearest match’ explanatory variables (probabilistic link)

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DAMES portal architecture overview



(Note: Security omitted)



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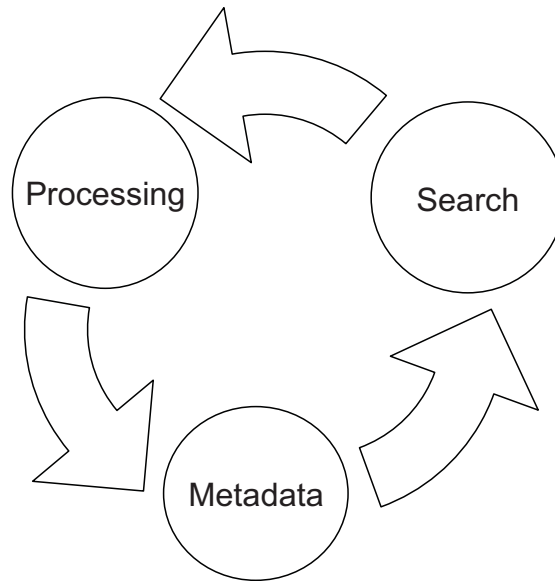
Key role for metadata

- *Metadata records are absolutely core to the functioning of the portal infrastructure*
 - For adequate, searchable records for the heterogeneous resources (data tables, command files, notes and documentation)
 - To connect the resources and the data management tools
 - To document the data sets resulting from application of the data management tools: inputs, process, rationale,...
- (See the metadata “cycle” on the next slide)



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The metadata "cycle"



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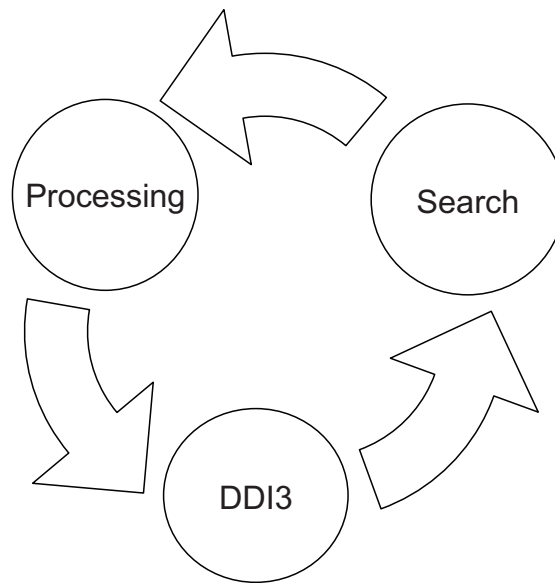
Choice of metadata standard

- DAMES requirements:
 - (Micro-)data based, very general, lifecycle oriented, Grid friendly
- Blum and Turner surveyed various metadata approaches
 - Common Warehouse Model: business-oriented, CORBA not grid (👎)
 - Dublin Core: too simplistic unextended (👎)
 - Encoded Archival Description: not data oriented (👎)
 - eGovernment Metadata Standard: not data oriented (👎)
 - MACHine Readable Cataloging: bibliographic (👎)
 - Metadata Encoding and Transmission Standard: DTD based, less flexible (👎)
 - MetaDater: Superseded (👎)
 - Open Archives Initiative: Related to Dublin Core (👎)
 - Statistical Data and Metadata Exchange: macro-data focussed (👎)
 - DDI 3: Just right! (👍)



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The metadata "cycle", again



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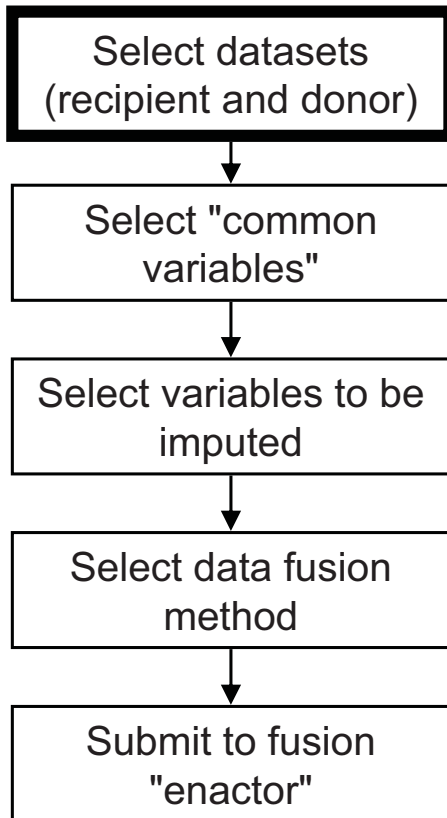
Example: The Fusion Tool

- Scenario: A soc sci researcher wishes to fuse Scottish Household Survey data with privately collected study data:
 - Uses the data curation tool to upload the data – the portal captures metadata
 - Uses the data fusion/imputation tool to select the data, identify corresponding variables, and to generate a derived dataset (held in the portal)
 - The metadata about this derived dataset is stored and (may be) made public through the portal
 - Another researcher can now search the portal (metadata) for SHS data and find the derived dataset
- *DAMES metadata handling must facilitate this process*



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The Fusion Tool at work



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DAMES Data Fusion Wizard
Stage 1: Dataset Selection

Please select a donor dataset and a recipient dataset.

Dataset Selection

Donor Dataset

1970 British Cohort Study

The 1970 British Cohort Study (BCS70) is a continuing, multi-disciplinary longitudinal study which takes as its subjects all those living in England, Scotland and Wales who were born in one particular week in April 1970.

Get Description

Recipient Dataset

Active People Survey

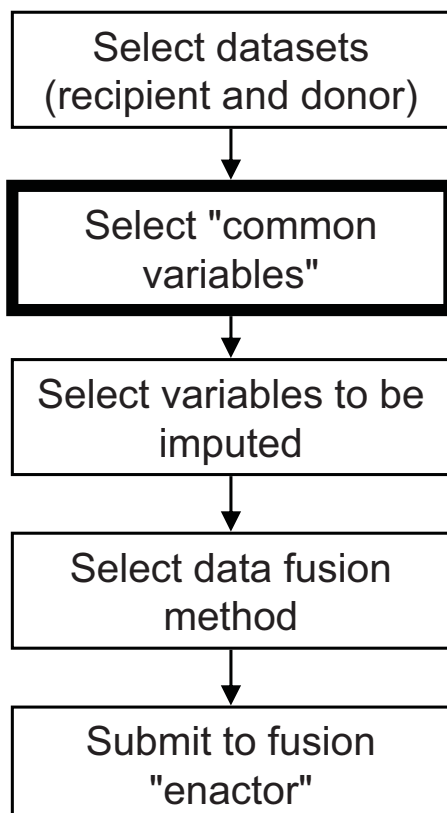
The Active People Survey is the largest ever survey of sport and active recreation to be undertaken in Europe.

Get Description

Stage 1 / 5 - Dataset Selection

Proceed ->

Metadata accessed



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DAMES Data Fusion Wizard
Stage 2: Common Variables Selection

Please select the variables that are common to the two datasets, select a mapping method, then click the "Add Selection" button. Multiple selections may be made. Click the "Proceed" button when all selections have been made.

Variable Selection

Donor Variable

A0004B

Get Description

Recipient Variable

KPI 1

Get Description

Mapping Method

Sirpenski Alpha Conjecture

Get Description

Add Selection

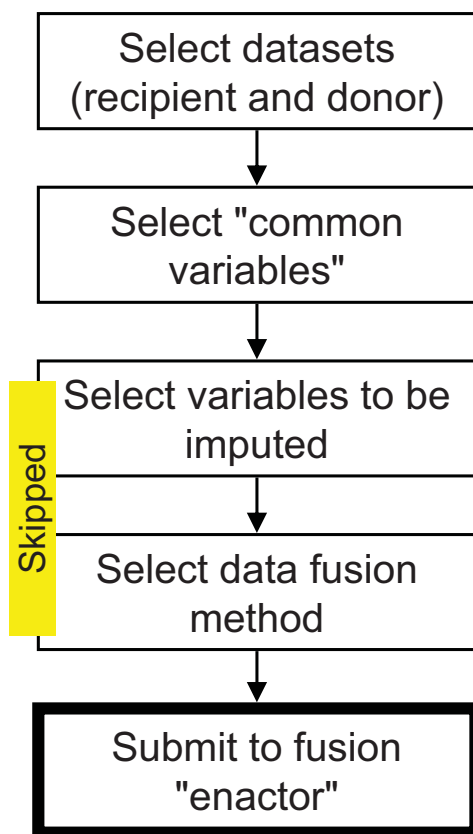
Stage 2 / 5 - Common Variables Selection

<- Back to dataset selection Proceed ->

Metadata accessed

✓ MAPPINGS: [DONOR: A0004B, RECIPIENT: KPI 1, METHOD: SIRPENSKI ALPHA CONJECTURE]





DAMES Data Fusion Wizard
Stage 5: Output

Below is a list of the fusion settings used and a link to the file. Please confirm your settings and click the submit button to submit your job to DAMES. You can monitor the status of your job and download the resultant file from the job monitoring portlet. Thank you for using the DAMES Data Fusion Tool.

Job Submission Details

Datasets

- Donor: 1970 British Cohort Study
- Recipient: Active People Survey

Mapped Variables

```
[<b>Donor:</b> A0004B,
<b>Recipient:</b> KPI 1,
<b>Mapping Method:</b> Sirpenski
Alpha Conjecture]
```

Imputed Donor Variables

```
[A0002]
[A0004B]
```

Data Fusion Method

- Jeber Fling Disambiguation

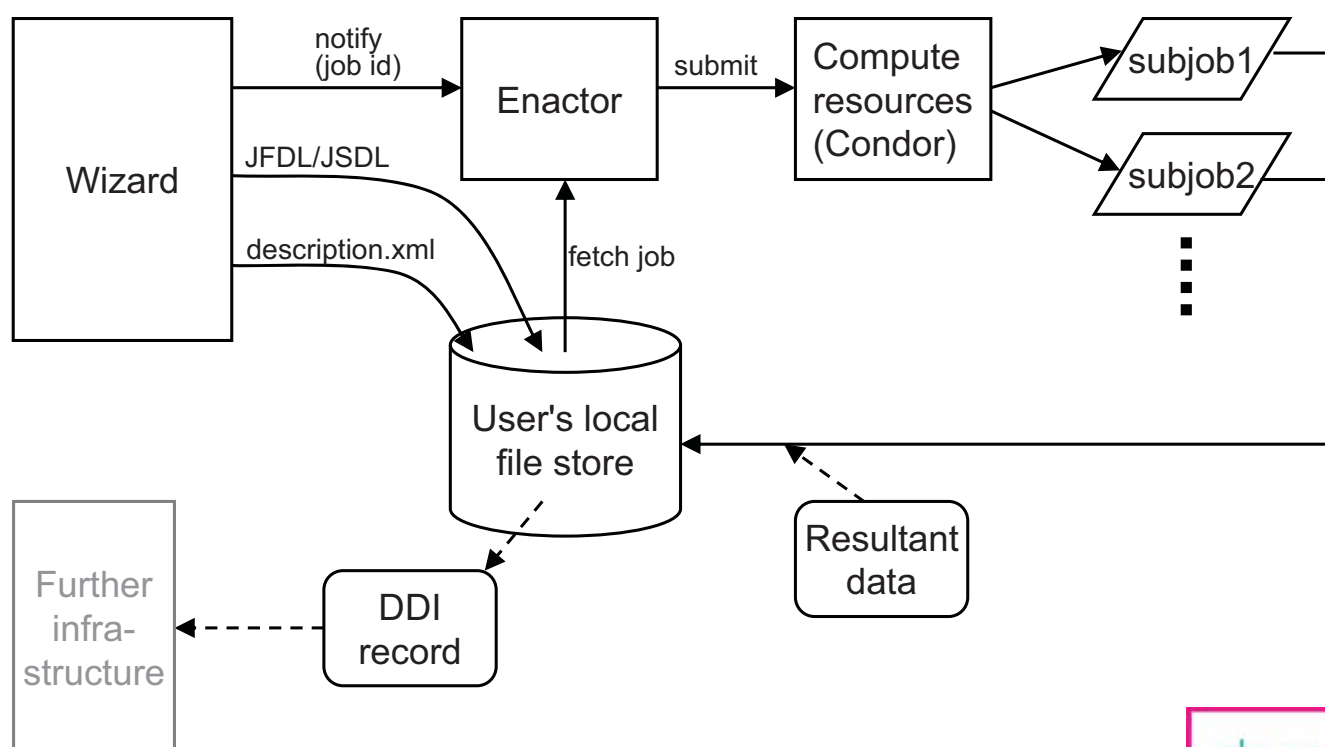
Step 5 / 5 - Output

<- Back to dataset selection Submit ->

Metadata for result dataset



Job submission: Information flow



The job flow description

- The fusion/imputation is potentially resource greedy
 - Generalise to using a pool of computing resources
- We use an extension of the Job Submission Description Language (JSDL) called JFDL (... Flow...) to submit the job to the computing resources pool
- The JFDL job description includes references to:
 - Input data sets
 - Processing steps and their relationships
 - Outputs
- This must be recorded in the DDI record for the resultant dataset (augmented with descriptive rationale)
 - We are automating the translation JFDL→DDI3



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JSDL/JFDL

```
<jSDL:Application>
  <jSDL:ApplicationName>DAMES::Fusion
  ...
  <jfdl:JobFlow>
    <jfdl:Dataset file="Gamma1.csv" id="acsv"/>
    ...
    <jfdl:Method submitFile="submit.imputeA" id="imputeA"/>
    <jfdl:Job id="j1">
      <jfdl:DataSequence>
        ...
        <jfdl:Method ref="imputeA"/>
      <jfdl:Job id="j3">
        <jfdl:Parent_job ref="j1"/>
        <jfdl:Parent_job ref="j2"/>
        ...
        <jfdl:Method ref="fusion"/>
      ...
    ...
  ...

```

A brief extract!



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The DDI3 metadata

A fragmentary outline!

<g:Concepts>

<!-- A conceptual component describing the data that are being fused is given for each mapped and imputed variable -->

...

<g:DataCollection>

<d:DataCollection>

<d:ProcessingEvent isIdentifiable="true" id="[Fusion Method Id]" ...>

<d:Coding isIdentifiable="true" id="[Fusion Method Submit File]">

<d:GenerationInstruction>

<d:SourceVariable ...> <r:URN... <d:Mnemonic>Donor Dataset

<d:SourceVariable ...> <r:URN... <d:Mnemonic>Recipient Dataset

...

<r:Command>

<r:CommandFile formalLanguage="[Language such as SPSS or STATA]">

...

<r:URI>[URI for the command file]

...

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Technology – other components

- Liferay portal
- eXist
 - XML based database – ideal for storing DDI3 metadata
- Condor
 - Job management
- iRODS
 - Highly flexible filestore
 - Capable of running automated processes on file upload: e.g. metadata extraction (e.g. STATA files), JFDL → DDI3 translation, & transfer from file store to metadata store

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Summary/Work in progress

- The DAMES infrastructure is still very much in the development phase
- We have a prototype outline portal, with prototype curation and fusion tools
- We are firmly fixed on using DDI 3 as our metadata standard, but we are still:
 - Refining the JFDL
 - Refining the DDI3
 - Improving generation of DDI3 from JFDL
 - Improving searching and discovery of datasets

Thank you!

